Comprehensive Guide to 3 PSTN Telephony Approaches for AI Calling

Detailed Architecture, Step-by-Step Code Integration, and Cost Analysis for Delivering AI Voice Directly into Customer Phone Ear Receivers

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1. Executive Summary & Hardware Realities

When placing automated AI sales calls, **Device B (the customer's phone) must hear the AI's synthesized voice crystal clear in their ear** without needing to install any application. Because physical Android smartphones physically block third-party apps from inserting digital audio files into standard SIM card phone calls, the industry standard solution is to connect your Python FastAPI backend directly to a **PSTN Telephony Infrastructure**.

2. Approach 1: Exotel Cloud Telephony Voice Stream API (Recommended for India)

Overview: Exotel is India's leading enterprise cloud communications platform. It provides direct Indian virtual numbers, low per-minute tariffs, and native bi-directional WebSockets (Exotel Voice Stream API).

```
[FastAPI Backend (app/services/call_dispatcher.py)] ■ 1. HTTP POST Call Request (Exotel REST API) ▼ [Exotel Cloud Telephony Engine] ■ 2. Dials Customer Phone Number via Indian Telecom Network ▼ [Customer's Mobile Phone] (Answers Call) ■ 3. Exotel Opens Bi-directional WebSocket to FastAPI ▼ [FastAPI /exotel/stream (WebSocket Receiver)] ■■■■ [Deepgram STT ➔ Gemini LLM ➔ ElevenLabs TTS]
```

How It Works technically:

- **Outbound Dialing**: Your backend calls Exotel's REST API `POST /v1/Accounts/{SID}/Calls/connect.json` specifying the customer's phone number.
- **WebSocket Handshake**: When the customer picks up, Exotel connects via WebSockets to `wss://your-domain.com/exotel/stream/{call_id}`.
- **Audio Exchange**: Exotel streams 8kHz mulaw audio chunks. Python FastAPI converts these to 16kHz PCM, passes them to Deepgram/Gemini, and returns the ElevenLabs AI voice back over WebSocket.

FastAPI Integration Code (app/api/exotel.py):

```
# app/api/exotel.py from fastapi import APIRouter, WebSocket, WebSocketDisconnect from app.services.call_orchestrator import CallOrchestrator router = APIRouter(prefix="/exotel", tags=["Exotel Telephony"]) @router.websocket("/stream/{call_id}") async def exotel_voice_stream(websocket: WebSocket, call_id: str): await websocket.accept() orchestrator = CallOrchestrator(call_id=call_id) await orchestrator.start() try: while True: audio_chunk = await websocket.receive_bytes() await orchestrator.process_audio_chunk(audio_chunk) except WebSocketDisconnect: await orchestrator.close()
```

3. Approach 2: Airtel Corporate SIP Trunking + FreeSWITCH / LiveKit Gateway

Overview: Bharti Airtel provisions a dedicated enterprise digital SIP channel trunk into your server infrastructure via a Session Border Controller (SBC). We host an open-source **FreeSWITCH Docker container** or **LiveKit SIP Gateway** to handle the telephony channel.

```
[FastAPI Backend (app/services/call_dispatcher.py)] ■ 1. Triggers ESL Command to FreeSWITCH Container ▼ [FreeSWITCH Docker / LiveKit SIP Gateway (SBC Container)] ■ 2. Dials Customer via Airtel Enterprise SIP Channel ▼ [Customer's Mobile Phone] (Answers Call) ■ 3. FreeSWITCH Bridges
```

```
Live RTP Audio Stream to Python AI Agent ▼ [Python AI Voice Agent (webrtc_agent.py)] ■■■■
[Deepgram STT → Gemini LLM → ElevenLabs/Aura TTS]
```

How It Works technically:

- **SIP Trunk Connection**: Airtel routes raw SIP signaling to your FreeSWITCH server IP over a dedicated IP pipe.
- **RTP Media Relay**: FreeSWITCH handles G.711 / G.722 RTP audio packet decoding natively and pipes the raw PCM stream directly into Python `webrtc_agent.py`.
- **Lowest Cost at Scale**: Since you own the FreeSWITCH container, call rates are just ■0.30/min flat rate with zero platform middleman fees.

FreeSWITCH Trigger Code (app/services/freeswitch_service.py):

```
# app/services/freeswitch_service.py import httpx async def trigger_airtel_sip_call(phone_number: str, call_id: str): url = "http://freeswitch-container:8081/api/originate" payload = { "gateway": "airtel_sip_trunk", "destination": phone_number, "call_id": call_id } async with httpx.AsyncClient() as client: res = await client.post(url, json=payload) return res.json()
```

4. Approach 3: Twilio Voice TwiML Media Streams

Overview: Twilio is the global leader in cloud communications. It provides worldwide phone numbers and TwiML WebSockets (``) to stream phone call audio directly into Python WebSockets.

```
[FastAPI Backend (app/services/call_dispatcher.py)] ■ 1. HTTP POST Call Request (Twilio REST API) ▼ [Twilio Global Voice Cloud] ■ 2. Dials Customer Phone Number ▼ [Customer's Mobile Phone] (Answers Call) ■ 3. Twilio Fetches TwiML & Opens WebSockets to FastAPI ▼ [FastAPI /twilio/stream (WebSocket Endpoint)] ■■■■ [Deepgram STT → Gemini LLM → ElevenLabs TTS]
```

How It Works technically:

- **TwiML Webhook**: Twilio calls your `/twilio/voice` HTTP endpoint, which responds with TwiML XML instructions to open a WebSockets media stream.
- **Audio Format**: Twilio streams base64-encoded 8kHz mulaw audio frames inside JSON payload messages.

Twilio Integration Code (app/api/twilio.py):

```
# app/api/twilio.py from fastapi import APIRouter, Response, WebSocket router = APIRouter(prefix="/twilio", tags=["Twilio Telephony"]) @router.post("/voice") async def twilio_voice(): twiml_xml = ''' ''' return Response(content=twiml_xml, media_type="application/xml")
```

5. Comprehensive Pricing & Feature Comparison Matrix

Cost modeled for **100 calls/day × 30 days = 3,000 calls / 12,000 call minutes per month**:

Feature / Metric	Approach 1: Exotel Cloud	Approach 2: Airtel SIP Trunk	Approach 3: Twilio Voice
Target Market	India (Native Cloud)	India Enterprise	Global / International
Telephony Rate / Min	■0.60 / min ($0.0071)	**■0.30 / min ($0.0036)**	■1.52 / min ($0.0180)
Monthly Infrastructure	■2,000 / mo	■2,500 / mo	■97 / mo
Total Telephony (12k Mins)	■9,200 INR ($109.50)	**■6,100 INR ($72.60)**	■18,241 INR ($217.15)
AI Pipeline (STT+LLM+TTS)	■10,634 INR ($126.60)	■10,634 INR ($126.60)	■10,634 INR ($126.60)

TOTAL MONTHLY COST	**■19,859 INR ($236.40)**	**■16,759 INR ($199.50)**	**■28,900 INR ($344.05)**
Effective Per Min Cost	**~■1.65 INR / min**	**~■1.39 INR / min**	**~■2.40 INR / min**
Setup Complexity	Low (Pure API)	Medium (FreeSWITCH)	Low (Pure API)

6. Final Recommendation

1. **Recommended Primary Option for Indian Market**: **Exotel Cloud Telephony**. It provides the ideal balance of zero telephony hardware maintenance, fast FastAPI WebSocket integration, and a low cost of **■1.65/min total**.
2. **Recommended Option for Maximum Scaling & Lowest Rate**: **Airtel Corporate SIP Trunk + FreeSWITCH** (**■1.39/min total**).